



Reconstructing the distribution of the non-native sea anemone, *Diadumene lineata* (Actiniaria), in the Canadian Maritimes: Local extinction in New Brunswick and no regional range expansion in Nova Scotia since its initial detection

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ABSTRACT

Since its first observations in New Brunswick (Sam Orr's Pond) and Nova Scotia (Bedford Institute of Oceanography in Dartmouth) in 2009 and 2013, respectively, *Diadumene lineata* has been observed spreading to new locations in the Canadian Maritimes. In 2017 and 2018, regional surveys for non-native species did not detect *D. lineata* in New Brunswick, suggesting local extinction. Recent surveys also did not detect *D. lineata* in Prince Edward Island. However, between 2013 and 2018, *D. lineata* was detected in Nova Scotia (in chronological order) from Lunenburg Harbour (Lunenburg), Hermans Island (Lunenburg), Black Rock Beach (Halifax), Oak Island Marina (Chester), and Dartmouth Yacht Club (Dartmouth). These new observations combine to reconstruct the known recent distribution of this non-native sea anemone in the Canadian Maritimes. Further, after five years since its first detection in Nova Scotia, the regional distribution of *D. lineata* remained unchanged, restricted to a region of approximately 100 km between Halifax and Lunenburg. Although the primary focus is on *D. lineata*, additional records of the non-native macroalga, *Codium fragile*, from Sam Orr's Pond and the native sea anemone, *Metridium senile*, fouling dock structures throughout the Canadian Maritimes are included in this report as *M. senile* commonly overlaps in habitat with *D. lineata*.

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1. Introduction

The orange-striped green sea anemone, *Diadumene lineata* (Verrill, 1869), is an infralittoral species considered to be native to China and Japan, where it was first described (Verrill, 1869; Uchida, 1932). It is now known as one of the most widely distributed sea anemones in the world, as it has been introduced via a variety of ways, including (a) the translocation of fouled boat hulls, bivalve aquaculture products, and aquaculture equipment, (b) ballast water exchange, and (c) long-distance rafting on anthropogenic debris (Verrill, 1898; Shick, 1976; Shick and Lamb, 1977; Gollasch and Riemann-Zürneck, 1996; Carlton et al., 2017).

Diadumene lineata is a relatively small, solitary sea anemone with a maximum diameter of up to 22 mm at the base, although

bodies ranging from 5 to 10 mm in diameter are typically encountered (Uchida, 1932; Hand, 1955; Fautin and Hand, 2007; Fautin et al., 2009). The smooth columnar body is greyish-green in colour with thin, orange or white longitudinal stripes (Uchida, 1932; Hand, 1955). In its native range, up to four colour morphs have been described, which vary in stripe size, colour, and arrangement (Uchida, 1936; Ryan and Kubota, 2016). Tentacles are slender and typically number about 25 but can be as many as 100 (Carlgren and Hedgepeth, 1952; Fautin and Hand, 2007). Individuals expel defensive acontia when disturbed (Uchida, 1932). A native congener, *Diadumene leucolea* (Verrill, 1866) is known from the Canadian Maritimes (Thomas et al., 1983); it can be distinguished from *D. lineata* by its transparent or pinkish body, large cnclides, and the absence of coloured stripes on the body column.

Diadumene lineata inhabits marine and estuarine environments tolerating a wide range of salinities (5 to 35 psu) and water temperatures (0 to 27.5 °C), which contributes to its status as a global invader (Shick, 1976; Seaton, 1985; Konecny and Harley,

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2019). Outside its native range, *D. lineata* mainly reproduces asexually either by longitudinal fission or pedal laceration (Shick, 1976; Ting and Geller, 2000). However, mixed sex populations containing eggs and sperm have been observed along the east coast and west coast of the United States (Newcomer et al., 2019; Ryan and Miller, 2019). The planula larvae have a disposition to attach to boat hulls that they encounter, suggesting that this is one of the main introduction vectors (Gollasch and Riemann-Zürneck, 1996).

Throughout the world, *D. lineata* and the sea anemone, *Metridium senile* (Linnaeus, 1761) are often found as part of the fouling community on artificial structures in marinas and harbours (Griffiths et al., 1996; Hancock et al., 2017). Both species have a cosmopolitan distribution but, at present, *D. lineata* has not been detected from southern Africa, Australia, the Arctic Ocean, and the Southern Ocean and *M. senile* has not been detected from Oceania, and the Southern Ocean (Hancock et al., 2017; OBIS, 2019). *Diadumene lineata* can be easily distinguished from *M. senile* based on external features: *M. senile* has a long, smooth columnar body with numerous slender tentacles, giving a feathery appearance in larger specimens. Its body colour is uniform and ranges from orange to pink to white (Fox and Pantin, 1941). The parapet, in the distal region of the column, typically forms a prominent collar-like ring (Batham and Pantin, 1951; Fautin and Hand, 2007).

In the Canadian Maritimes, *D. lineata* was first detected in October 2009 as a non-native species at Sam Orr's Pond, New Brunswick (Saunders et al., 2013). In November 2013, *D. lineata* was documented for the second time in the Canadian Maritimes off a dock at the Bedford Institute of Oceanography in Dartmouth, Nova Scotia (Moore et al., 2014).

This study consolidates several regional surveys that detected *D. lineata* since the records of Saunders et al. (2013) and Moore et al. (2014) to better understand its distribution in the Canadian Maritimes (New Brunswick, Nova Scotia, and Prince Edward Island). Here, observations of *D. lineata* from five sources are documented: (a) 2013 benthic community surveys in Lunenburg, Nova Scotia, (b) 2017 surveys throughout New Brunswick targeting *D. lineata*, (c) a 2017 survey of Sam Orr's Pond, New Brunswick targeting *D. lineata* and another non-native species, *Codium fragile* (Suringar) Hariot, 1889, (d) a 2017 observation from Halifax, Nova Scotia, contributed by a citizen scientist, and (e) 2018 surveys throughout the Canadian Maritimes targeting *D. lineata* and *M. senile*.

2. Methods

2.1. 2013

2.1.1. Artificial structures

In Nova Scotia, sampling of benthic species was done at the Lunenburg Yacht Club on Hermans Island (44.4109°N; 64.3218°W), and along two wooden wharves in Lunenburg Harbour (44.3755°N; 64.3095°W). Settlement plates were deployed at random locations within each site, suspended from floating docks (N = 50 plates in Hermans Island and N = 6 plates in Lunenburg Harbour) on 20 July and retrieved on 19 September. Between 18 and 19 September, the same number of quadrats per site were scraped by SCUBA divers from the undersides of floating docks and the vertical sides of wooden pilings. The dimensions of the quadrats and settlement plates both measured 20 × 20 cm in size and, upon collection, benthic samples were transported to a laboratory where they were maintained in flowing seawater holding tanks until taxonomic examination (see Section 2.4 below). Only observations of *D. lineata* and *M. senile* from these settlement plates and benthic quadrats were used for this report.

2.2. 2017

2.2.1. Artificial structures

At each of the 20 sites throughout New Brunswick (Table 1), a systematic survey of the undersides of floating docks as viewed from the water surface at the edge of floating docks was done between 1 and 9 October. This site-level survey was followed by a detailed examination of 20 different 1-m wide dock sections for the presence and absence of *D. lineata*. Only the presence of *M. senile* was noted from these sites in New Brunswick (i.e., no data on their absences).

2.2.2. Re-survey of Sam Orr's Pond

As the original site of the initial reports of *D. lineata* in the Canadian Maritimes and *C. fragile* in the southwest New Brunswick region of the Bay of Fundy, the shallow waters of Sam Orr's Pond (45.1583°N; 67.0442°W) were re-surveyed for both non-native species during low tide on 19 October 2017.

2.2.3. iNaturalist species record

On 6 October 2017, an observation of a sea anemone (later identified as *D. lineata*) attached to the chela of *Cancer borealis* Stimpson, 1859 was made by a citizen scientist at Black Rock Beach, Point Pleasant Park in Halifax, Nova Scotia (44.6247°N; 63.5640°W). The observation, accompanied by a photograph of the specimen, was uploaded to iNaturalist (<https://www.inaturalist.org/>), which is an online database of species records and social network to crowdsource taxonomic identifications of these records. The iNaturalist network and other similar citizen science databases were further searched for potential photographic observations of the species in the Canadian Maritimes.

2.3. 2018

2.3.1. Artificial structures

Surveys of the undersides of floating docks as viewed from the water surface were done at six sites in New Brunswick, 21 sites in Nova Scotia, and eight sites in Prince Edward Island for the occurrence and abundance of two species of sea anemones, *D. lineata* and *M. senile*, between 28 September and 7 October 2018 (Table 1). Sea anemone abundance was classified using the semi-quantitative SACFOR scale, which consist of six categories ranging from super-abundant to rare (Connor and Hiscock, 1996).

2.4. Morphological identification

The identities of *D. lineata* and *M. senile* specimens collected live (2013 survey) and those observed and photographed *in situ* (2017 and 2018 survey) were confirmed by examination of external morphological features, the method also used by Griffiths et al. (1996), Saunders et al. (2013), Moore et al. (2014), and Hancock et al. (2017). Similarly, the identities of specimens of *M. senile* observed *in situ* were based on external features. The macroalga *C. fragile* was identified morphologically by its spongy, cylindrical, finger-like, green thalli that branches dichotomously (Hubbard and Garbary, 2002).

3. Results

3.1. New Brunswick

No new records of *D. lineata* were made as a result of the 2017 and 2018 surveys of submerged portions of dock structures (Table 1). In Sam Orr's Pond, no new observations of *D. lineata* were made (Table 1); however, *C. fragile* continued to be present at this site as of October of 2017 (Fig. 1).

Table 1Presence and absence of the non-native sea anemone, *Diadumene lineata*, and the native sea anemone, *Metridium senile*, from the Canadian Maritimes.

Site	Survey date(s)	Latitude (°)	Longitude (°)	Occurrence of <i>D. lineata</i>	Occurrence of <i>M. senile</i>	Observer ^a
New Brunswick						
Sam Orr's Pond	14 October 2009	45.1620	−67.0450	Present	No data	Saunders et al. (2013)
Shediac	01 October 2017	46.2274	−64.5448	Not detected	No data	KCKM
L'Etete	04 October 2017	45.0513	−66.8954	Not detected	No data	KCKM
St. Andrews Harbour	04 October 2017	45.0714	−67.0545	Not detected	Present	KCKM
Blacks Harbour	05 October 2017	45.0558	−66.7943	Not detected	Present	KCKM
Ingalls Head	05 October 2017	44.6626	−66.7563	Not detected	No data	KCKM
North Head	05 October 2017	44.7630	−66.7494	Not detected	No data	KCKM
Back Bay	06 October 2017	45.0559	−66.8632	Not detected	No data	KCKM
Beaver Harbour	06 October 2017	45.0688	−66.7387	Not detected	No data	KCKM
Boynes Cove	06 October 2017	45.1309	−66.4979	Not detected	No data	KCKM
Chance Harbour	06 October 2017	45.1226	−66.3514	Not detected	No data	KCKM
Fairhaven	06 October 2017	44.9640	−67.0080	Not detected	No data	KCKM
Lorneville	06 October 2017	45.1923	−66.1486	Not detected	No data	KCKM
Leonardville	07 October 2017	44.9717	−66.9527	Not detected	No data	KCKM
Lords Cove	07 October 2017	45.0060	−66.9466	Not detected	No data	KCKM
Seal Cove	07 October 2017	44.6491	−66.8385	Not detected	No data	KCKM
Stuart Town	07 October 2017	45.0190	−66.9377	Not detected	No data	KCKM
Dipper Harbour	08 October 2017	45.0943	−66.4180	Not detected	No data	KCKM
Five Fathom Hole	08 October 2017	45.1867	−66.2575	Not detected	No data	KCKM
Saint John	08 October 2017	45.2723	−66.0657	Not detected	No data	KCKM
St. Andrews Biological Station	09 October 2017	45.0825	−67.0848	Not detected	No data	KCKM
Sam Orr's Pond	19 October 2017	45.1583	−67.0442	Not detected	No data	KCKM
Richibucto	29 September 2018	46.6791	−64.8617	Not detected	Not detected	HEG
St. Andrews Harbour	30 September 2018	45.0715	−67.0544	Not detected	Present (abundant)	HEG
Beaver Harbour	01 October 2018	45.0684	−66.7385	Not detected	Present (abundant)	HEG
Dipper Harbour	01 October 2018	45.0938	−66.4180	Not detected	Present (abundant)	HEG
Seeleys Cove	01 October 2018	45.0938	−66.6355	Not detected	Not detected	HEG
Stuart Town	01 October 2018	45.0189	−66.9374	Not detected	Present (abundant)	HEG
Nova Scotia						
Hermans Island (quadrats)	18 September 2013	44.4109	−64.3218	Present (28% of quadrats)	Present (16% of quadrats)	HLH
Hermans Island (settlement plates)	20 July – 19 September 2013	44.4109	−64.3218	Not detected	Not detected	HLH
Lunenburg Harbour (quadrat)	19 September 2013	44.3755	−64.3095	Present (17% of quadrats)	Not detected	HLH
Lunenburg Harbour (settlement plates)	20 July – 19 September 2013	44.3755	−64.3095	Present (17% of plates)	Not detected	HLH
Bedford Institute of Oceanography	19 November 2013	44.6810	−63.6101	Present	No data	Moore et al. (2014)
Black Rock Beach	06 October 2017	44.6247	−63.5640	Present	No data	CNC
Brooklyn	02 October 2018	44.0505	−64.6898	Not detected	Not detected	HEG
Dennis Point	02 October 2018	43.6181	−65.7888	Not detected	Present (occasional)	HEG
Digby	02 October 2018	44.6267	−65.7539	Not detected	Present (rare)	HEG
Meteghan	02 October 2018	44.1939	−66.1668	Not detected	Not detected	HEG
Yarmouth	02 October 2018	43.8315	−66.1230	Not detected	Not detected	HEG
Armdale Yacht Club	03 October 2018	44.6350	−63.6137	Not detected	Present (abundant)	HEG
Gold River	03 October 2018	44.5470	−64.3166	Not detected	Not detected	HEG
Hubbards	03 October 2018	44.6379	−64.0592	Not detected	Not detected	HEG
Oak Island Marina	03 October 2018	44.5156	−64.2919	Present (super-abundant)	Present (rare)	HEG
Purcell's Cove	03 October 2018	44.6108	−63.5736	Not detected	Not detected	HEG
Shearwater	03 October 2018	44.6280	−63.5201	Not detected	Not detected	HEG
St. Margaret's Bay	03 October 2018	44.6354	−63.9191	Not detected	Present (occasional)	HEG
Bedford	04 October 2018	44.7257	−63.6637	Not detected	Not detected	HEG
Dartmouth Yacht Club	04 October 2018	44.7007	−63.6113	Present (abundant)	Present (abundant)	HEG
Guysborough	04 October 2018	45.3920	−61.4971	Not detected	Present (abundant)	HEG
North Sydney	04 October 2018	46.1918	−60.2684	Not detected	Present (abundant)	HEG
Baddeck	05 October 2018	46.0995	−60.7483	Not detected	Present (abundant)	HEG
Inverness	05 October 2018	46.2302	−61.3166	Not detected	Not detected	HEG
Pictou	05 October 2018	45.6745	−62.7117	Not detected	Not detected	HEG
Port Hawkesbury	05 October 2018	45.6138	−61.3651	Not detected	Present (abundant)	HEG
Pugwash	05 October 2018	45.8465	−63.6632	Not detected	Present (common)	HEG
Prince Edward Island						
Borden-Carleton	06 October 2018	46.2494	−63.7012	Not detected	Not detected	HEG
Charlottetown	06 October 2018	46.2312	−63.1213	Not detected	Present (abundant)	HEG
French River	06 October 2018	46.5089	−63.5006	Not detected	Not detected	HEG
Georgetown	06 October 2018	46.1789	−62.5305	Not detected	Present (common)	HEG
Malpeque-Darnley	06 October 2018	46.5344	−63.6645	Not detected	Not detected	HEG
Stanley	06 October 2018	46.4640	−63.4591	Not detected	Not detected	HEG
Summerside	06 October 2018	46.3885	−63.7857	Not detected	Not detected	HEG
Tracadie Bay	06 October 2018	46.4029	−63.0296	Not detected	Not detected	HEG

^aCNC = C.N. Chapman; HEG = H.E. Glon; HLH = H.L. Hawk; KCKM = K.C.K. Ma.

3.2. Nova Scotia

In Lunenburg, new occurrences of *D. lineata* were observed in 2013 in Lunenburg Harbour and Hermans Island (Table 1; Fig. 2). In September of 2013, a total of 37 individuals of *D. lineata* were collected in quadrats scraped from the undersides of floating docks at Hermans Island (18.5 individuals m^{-2}), one individual in a quadrat scraped from a wooden piling in Lunenburg Harbour (4.2 individuals m^{-2}), and one individual on a settlement plate in Lunenburg Harbour (4.2 individuals m^{-2} ; Fig. 1). *Diadumene lineata* was not observed on settlement plates deployed at Hermans Island. Twenty-two individuals of *M. senile* were observed in benthic quadrats at Hermans Island (11.0 individuals m^{-2}). Out of the 50 quadrats at Hermans Island, three had both *D. lineata* and *M. senile*. *Metridium senile* was not detected on settlement plates at Hermans Island, nor was it detected by either sampling method in Lunenburg Harbour.

In October of 2017, a new observation of an individual of a sea anemone was made at Black Rock Beach in Halifax (Fig. 2). This observation (and accompanying photograph; Fig. 1) was uploaded to iNaturalist and later identified as *D. lineata* (Table 1). Further database enquiries did not yield additional observations of the species in the Canadian Maritimes.

As part of the survey of floating docks in October of 2018, two new occurrences of *D. lineata* (Fig. 1) were reported from Oak Island Marina near Chester (abundance: super-abundant; Fig. 2) and Dartmouth Yacht Club (abundant with two colour morphs present; Fig. 2) where populations of *D. lineata* co-occurred with *M. senile* (Table 1 and Fig. 3).

3.3. Prince Edward Island

Diadumene lineata was not detected from Prince Edward Island during the 2018 survey.

4. Discussion

4.1. Overall distribution in the Canadian Maritimes

The lack of observations of *D. lineata* in New Brunswick from the combined surveys of 2017 and 2018 suggest that this species went extinct (permanently or temporarily) in New Brunswick at some time between 2009 and 2017. The systematic surveys in 2017 and 2018 show that its distribution in the Canadian Maritimes is restricted to a region of approximately 100 km from Halifax to Lunenburg, Nova Scotia (Fig. 2). At some of the sites (i.e., Dartmouth Yacht Club, Hermans Island, and Oak Island Marina), *D. lineata* co-occurred in close proximity with *M. senile*.

4.2. Local extinction in New Brunswick

The local extinction of *D. lineata* from southwest New Brunswick is not unique and follows a similar extinction pattern in the Gulf of Mexico and in Germany. In the Gulf of Mexico, it was first observed from Port Aransas in 1952 but it was not recorded again from this site despite regular sampling in the 1980s (Carlgren and Hedgepeth, 1952; Hancock et al., 2017). In Germany, it was detected once in the Büsum (on the North Sea) between 1920 and 1924 (Gollasch and Riemann-Zürneck, 1996). Such patterns of local extinctions of *D. lineata* are known from the literature and could be caused by an environmental catastrophe exerted on a small, non-native population consisting of a single clone (Shick et al., 1979; Gollasch and Riemann-Zürneck, 1996; Hancock et al., 2017).

The likelihood of future re-introduction and establishment of an introduced species in a region can be difficult to determine,

even in the context of climate change. For example, in the Gulf of Mexico, *D. lineata* was detected for the second time on a jetty on Galveston Island, Texas, in November of 2016, about 280 km north of Port Aransas and sixty-four years after its initial observations (Hancock et al., 2017), which suggests a separate introduction event. Correspondingly, the local re-appearance of *D. lineata* in New Brunswick is also possible. Reporting local extinctions of non-native populations is an important endeavour (e.g., see Robinson et al., 2005), which should help focus finite resources on monitoring existing biological invasions while working to predict future re-introduction events.

In contrast with the extinction of *D. lineata* from southwest New Brunswick, the non-native macroalga *C. fragile* was still present as of October 2017 in Sam Orr's Pond, which was the site of its first observation from the Bay of Fundy made on 23 July 2008 (Saunders et al., 2013). Since it was first documented in Atlantic Canada in December 1991, *C. fragile* is now a well-established and widely distributed throughout the Atlantic coast of Nova Scotia and the Gulf of St. Lawrence, with northernmost records from substantially colder waters of Newfoundland (Bird et al., 1993; Garbary et al., 1997; Schmidt and Scheibling, 2007; Matheson et al., 2014). However, the geographic extent of *C. fragile* in the Bay of Fundy remains unknown, since all observations of this species were reported from the one site in Sam Orr's Pond. Therefore, systematic surveys for *C. fragile* of the Bay of Fundy should continue to be carried out to determine its distribution and to monitor its spread and impact.

4.3. No regional range expansion in Nova Scotia

The seasonal timing of the *D. lineata* observations in Lunenburg (September 2013) was approximately two months earlier than the previously documented observation in Dartmouth (November 2013; Moore et al., 2014). By considering a recruitment and growth period of as little as three weeks and up to two months to obtain the sizes observed (at least 7 mm in basal diameter), these new records from Lunenburg suggest that this species was distributed from Halifax to Lunenburg at least as early as the summer of 2013.

The first report of *D. lineata* from Nova Scotia was based on an individual collected from a settlement plate. However, this study found that detection of *D. lineata* in the region was more effective using surveys of established communities on floating docks and wharves than by using settlement plates. The low recruitment of *D. lineata* on plates despite its abundance on anthropogenic structures at Lunenburg Harbour and Hermans Island sites indicated that either (a) the timing of reproduction was not during late summer or (b) the species does not often reproduce sexually in the region, as has been documented in other regions of its introduced range (Shick, 1976; Shick and Lamb, 1977; Dunn, 1982; Fukui, 1991; Ting and Geller, 2000; Ryan and Miller, 2019; Newcomer et al., 2019).

The reconstructed range (from Halifax to Lunenburg) suggests that the regional distribution of *D. lineata* in Nova Scotia has remained relatively unchanged in the five years after its initial report. This lack of range expansion along the south shore Nova Scotia after five years is surprising given that its status as a globally successful invader (Seaton, 1985) and that the region is highly connected by commercial ships and recreational boats (Simard et al., 2017). Remarkably, extensive government monitoring for aquatic invasive species from 2012 to 2015 throughout New Brunswick and Nova Scotia did not mention any observations of *D. lineata* (DFO, 2015; Sephton et al., 2017). The probability of detection also depends on factors such as species density, sampling effort, method, and season. The high abundances of

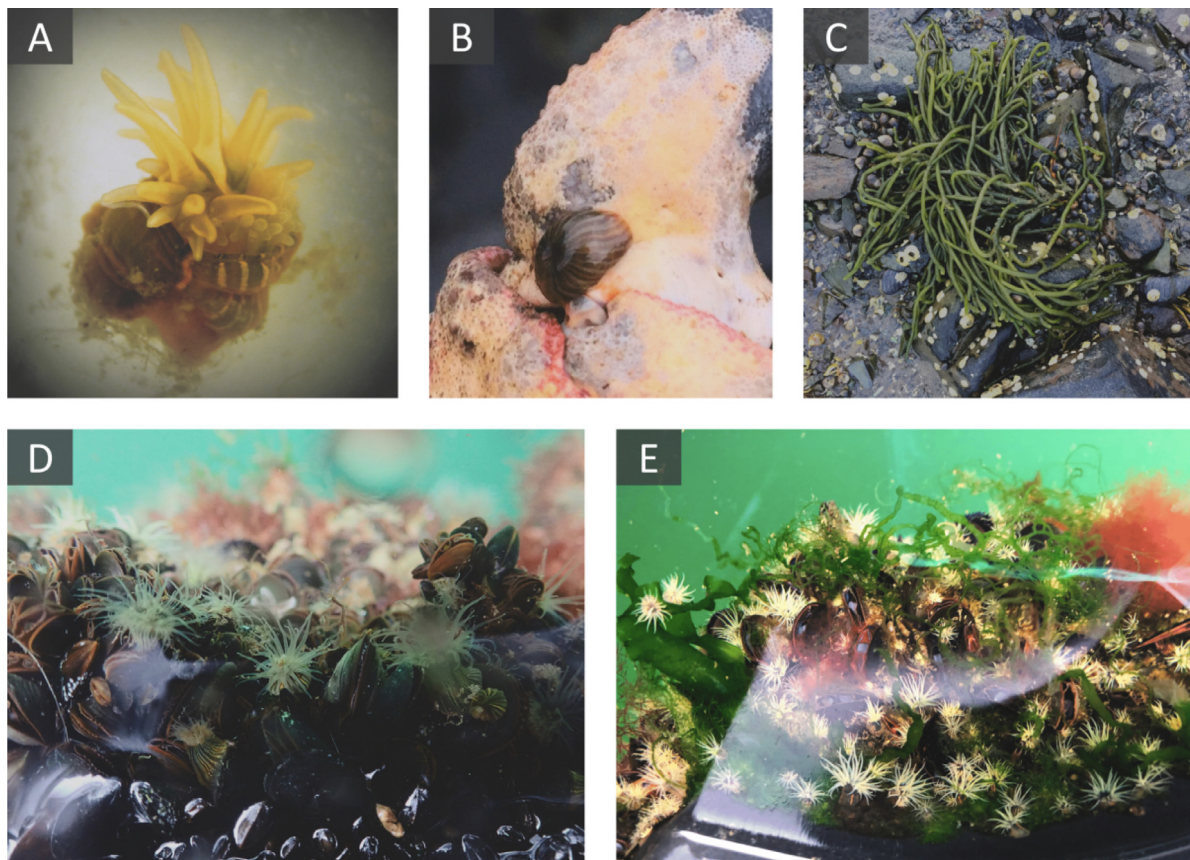


Fig. 1. Observations of *Diadumene lineata* in the Canadian Maritimes and *Codium fragile* from the Bay of Fundy. (A) A slightly contracted specimen of *D. lineata* collected from Hermans Island, Nova Scotia, in 2013 (basal diameter = 7 mm); (B) a completely contracted specimen of *D. lineata* observed from Black Rock Beach, Nova Scotia, in 2017; (C) *C. fragile* from Sam Orr's Pond, New Brunswick, in 2017; (D) extended specimens of *D. lineata* from Dartmouth Yacht Club, Nova Scotia, in 2018; (E) extended specimens *D. lineata* from Oak Island Marina, Nova Scotia, in 2018. Scale bars not available. Photo credit: H.L. Hawk (A), C.N. Chapman (B), K.C.K. Ma (C), H.E. Glon (D, E).

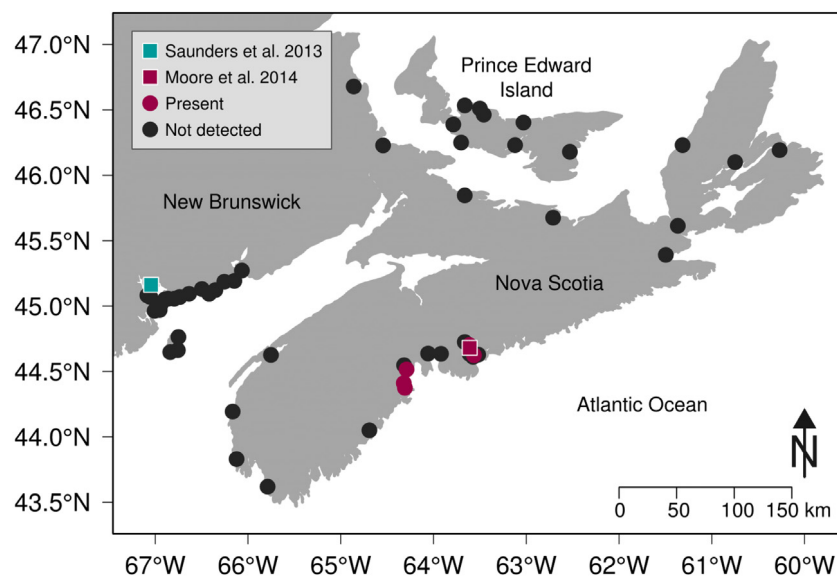


Fig. 2. Distribution *Diadumene lineata* in the Canadian Maritimes based on occurrence records collated in the present study and records documented by [Saunders et al. \(2013\)](#) in New Brunswick and by [Moore et al. \(2014\)](#) in Nova Scotia.

D. lineata observed at some sites that are also regularly surveyed by the government (e.g., Lunenburg) indicate that this non-native sea anemone is not targeted for monitoring. The inclusion

of this species in regular monitoring can substantially improve uncertainties associated with any absence records. The collection and identification of the iNaturalist observation of *D. lineata*

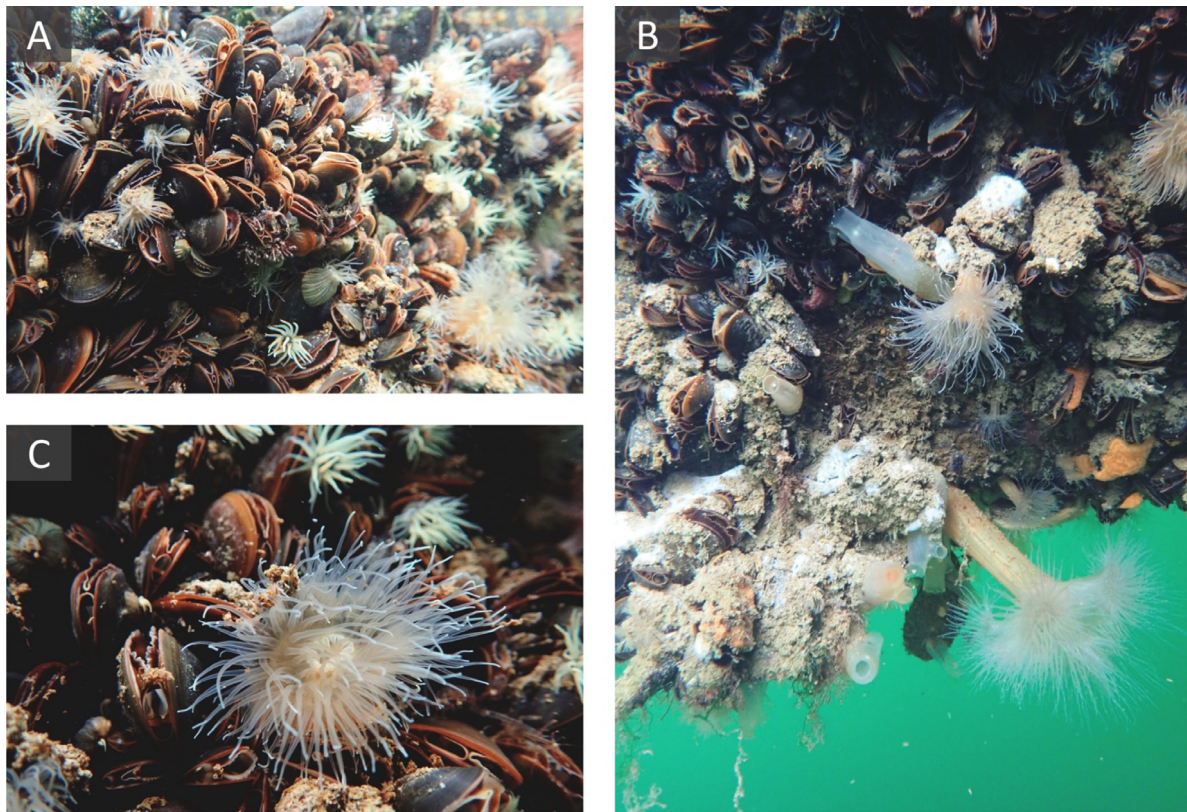


Fig. 3. Two sea anemones, the non-native *Diadumene lineata* and the native *Metridium senile*, co-occurring on fouling dock structures in Dartmouth Yacht Club, Nova Scotia. (A) Oblique view of the fouling community; (B) lateral view of the body of *M. senile*; (C) oral disc view of *M. senile*. Scale bars not available. Photo credit: H.E. Glon.

from Black Rock Beach in Halifax highlights the potential role of citizen-science networks for detecting non-native species and monitoring their distribution in the Canadian Maritimes.

4.4. Conclusions

In this report, the known distribution of *D. lineata* in the shallow waters of the Canadian Maritimes is described. In Nova Scotia, its occurrence is closely associated with harbours and marinas. This association suggests that human activities likely facilitate the transport and introduction of *D. lineata* along the south shore of Nova Scotia. The regional and international connectivity of the Canadian Maritimes via human activities increases the risk of re-introduction and further geographic spread in the future. Therefore, systematic monitoring across the region (e.g., [Sephton et al., 2017](#)) should be carried out regularly to track the progression of its geographic spread over time.

Documenting the occurrence and circumstances (e.g., the various methods leading to a confirmed observation) of detecting non-native species throughout their invasion process (from introduction to establishment and to regional spread) is important to reconstruct their invasion history. In particular, this report provides evidence that *D. lineata* have gone locally extinct in New Brunswick and that its distribution in Nova Scotia have remained unchanged after five years since 2013, when it was first detected in Nova Scotia.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Kevin C.K. Ma: Conceptualization, Methodology, Investigation, Resources, Data curation, Writing - original draft, Writing - review & editing, Visualization, Funding acquisition. **Heather E. Glon:** Conceptualization, Methodology, Investigation, Resources, Data curation, Writing - review & editing, Funding acquisition. **Heather L. Hawk:** Conceptualization, Methodology, Investigation, Resources, Data curation, Writing - review & editing, Funding acquisition. **Cody N. Chapman:** Investigation.

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